

Practical No. 9

Aim: Write a Java program that demonstrates array operations such as initialization, accessing elements, and performing basic array manipulations.

Source-Code:

```
public class ArrayOperations {
    public static void main(String[] args) {
        int[] numbers = {10, 20, 30, 40, 50};
        System.out.println("First element: " + numbers[0]);
        System.out.println("Third element: " + numbers[2]);

        System.out.println("Updated second element: " + numbers[1]);

        System.out.println("\nAll elements in the array:");
        for (int i = 0; i < numbers.length; i++) {
            System.out.println(numbers[i]);
        }

        int sum = 0;
        for (int i = 0; i < numbers.length; i++) {
            sum += numbers[i];
        }
        System.out.println("\nSum of array elements: " + sum);
        int max = numbers[0];
        for (int i = 1; i < numbers.length; i++) {
            if (numbers[i] > max) {
                max = numbers[i];
            }
        }
        System.out.println("Largest element: " + max);
    }
}
```

Output:

```
First element: 10
Third element: 30
Updated second element: 20

All elements in the array:
10
20
30
40
50

Sum of array elements: 150
Largest element: 50
```